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JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere

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An Autonomous Institute, with NAAC "A" Grade

Department of CSE\AI

Education to Eternity

"A place to learn; A chance to grow"

Session :-2025-26

MSE-II EXAMINATION

Course: B. Tech. in CSE \ AI

Subject: Multimodelling Development Systems

Year/Semester: 6th Semester (Third Year)

Max Marks: 20

Subject Code: CS6M006\AI6M006

Date: 08 / 04 /2026

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Understand the principles and practices of scalable systems for data science.
2. Apply skills for designing and implementing distributed computing systems.
3. Analyse the cloud computing technologies for data science.
4. Design and develop applications on Big Data platforms and their optimisations on commodity clusters and Clouds.
5. Implement data science algorithms and analytics using Big Data platforms.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Simulink.	L1/CO3	1
Q.2	What is sensitivity analysis.	L1/CO4	1
Q.3	State equation-based modelling.	L2/CO3	1
Q.4	Identify trade-off in multi-objective optimization.	L2/CO4	1

SECTION B

1. **All Questions are compulsory.**
2. **Solve answer in brief (two, three Sentence)**

Q. No.	Questions	Level/CO Mapped	Marks
Q.5	Interpret the concept of scenario-based simulation.	L3/CO4	2
Q.6	Write short note on simulation management and monitoring.	L2/CO3	2
Q.7	Explain the role of objective functions in optimization.	L2/CO4	2
Q.8	Describe the concept of multi-domain modelling in Modelica.	L2/CO3	2

SECTION C

1. **Solve any two questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.9	Describe modelling libraries and explain how component reuse improves system design.	L4/CO3	4
Q.10	Discuss the parameter estimation and calibration techniques in detail.	L3/CO4	4
Q.11	Explain the integration of heterogeneous models with suitable examples.	L2/CO3	4

All The Bests